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A Probing Model Training Settings

For all probing experiments, we train the probing model with an L_1 norm regularisation. The training objective is $\mathcal{L} = -\log P(y = y_i) + \lambda \|W\|_1$, where we set λ to 3×10^{-4} and y_i is the label. We train 20 times with different random seeds for each probing task, and we report the average and deviation in our experiments. We split the training and test datasets for the probing tasks, ensuring no overlapping questions between them.

B More Experimental Results on Knowledge Conflict Probing

We present the knowledge conflict probing results on Macnoise, NQSwap, ConflictQA using Llama2-7B in Figure 4, Figure 5 and Figure 6. The results match the trend discussed in Section 4, where the model exhibits an internal mechanism for identifying conflicts. The signal of knowledge conflict peaks around the 13th to 14th layers and gradually decreases in the later layers.

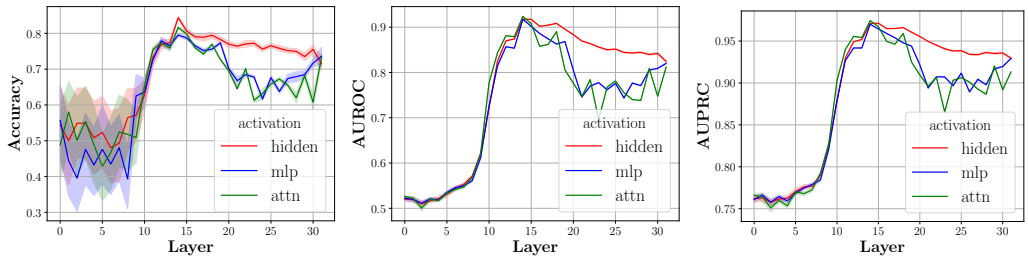

Figure 4: Knowledge conflict probing results using Llama2-7B on NQSwap.

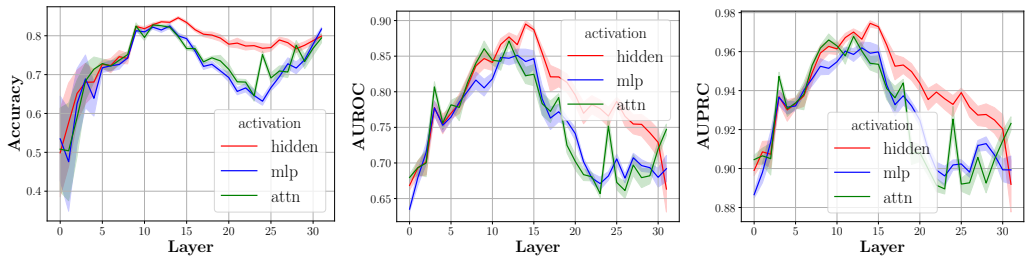

Figure 5: Knowledge conflict probing results using Llama2-7B on Macnoise.

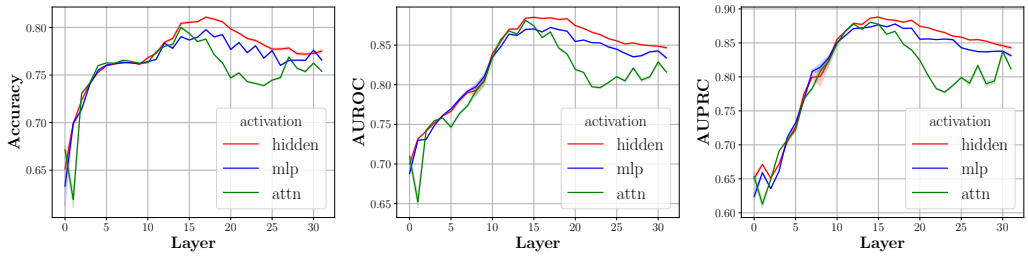

Figure 6: Knowledge conflict probing results using Llama2-7B on ConflictQA.

C More Analysis of Skewness Patterns of Residual Streams

We present the skewness of the hidden state of Llama2-7B on NQSwap in Figure 7. It shows the same pattern as we discussed in Figure 2, where the residual stream exhibits significantly more skewed distribution when using contextual knowledge compared with using parametric knowledge from the 17th layer.

In addition to NQSwap, we analyse the skewness pattern using Macnoise [Hong et al., 2024] and ConflictQA [Xie et al., 2024]. As shown in Figure 8, Figure 9, Figure 10, we find that the model also shows a similar skewness pattern with NQSwap, where the residual stream exhibits a more skewed distribution from middle layers when the model uses the contextual knowledge.

We also analyse the skewness of MLP and Self-Attention activations, presented in Figure 11, Figure 12, Figure 13, and Figure 14. However, we do not observe a specific skewness pattern in MLP and Self-Attention activations.

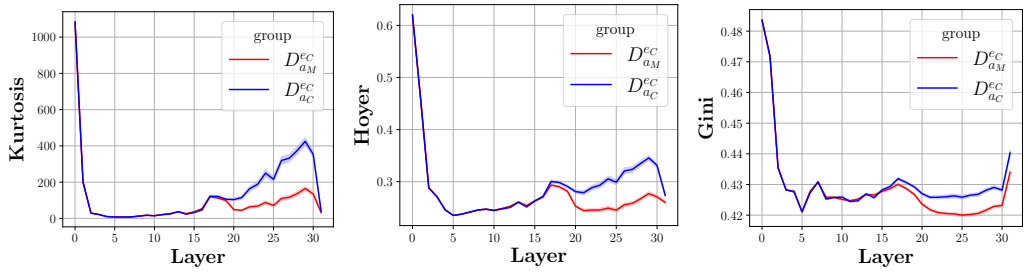

Figure 7: Skewness of the hidden states of Llama2-7B on NQSwap.

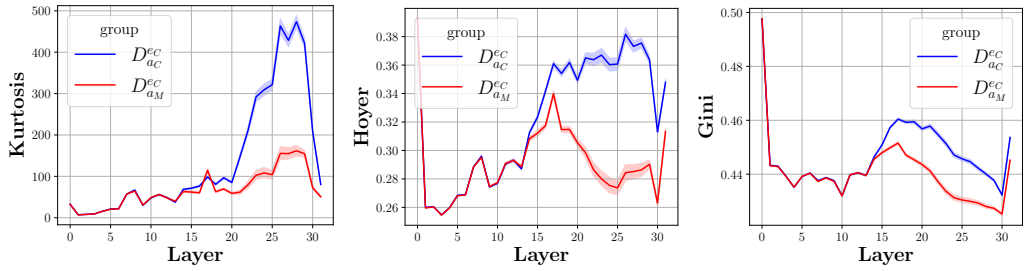

Figure 8: Skewness of the hidden states of Llama3-8B on Macnoise.

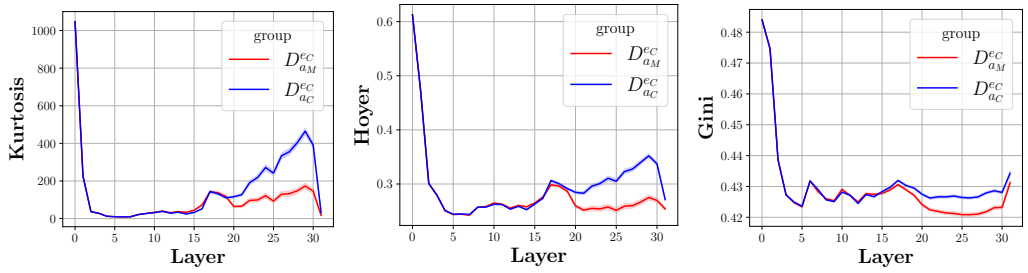

Figure 9: Skewness of the hidden states of Llama2-7B on Macnoise.